

LAXMINARAYAN SAHU

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EDUCATION

National Institute of Technology Karnataka, Surathkal

B.Tech in Information Technology; CGPA: 5.41

Aug. 2023 – May 2027 (Expected)

Surathkal, India

EXPERIENCE

SPriT Lab, IIT Jodhpur

Summer Research Intern — AI Engineering

May 2026 – Aug. 2026

Jodhpur, India

- Developed Omniac RAG, a Python retrieval-augmented generation (RAG) workflow that searches unstructured text and generates answers supported by source citations.
- Combined 2 retrieval methods, BM25 and dense search, using reciprocal rank fusion and neural reranking to select relevant evidence for user queries.
- Implemented natural language processing (NLP) claim verification and graph-based consensus to identify conflicting evidence, with query screening for suspicious inputs.
- Applied prompt engineering to assemble verified evidence and citation instructions for Ollama large language models (LLMs), with answer verification and abstention when evidence is insufficient.

PROJECTS

Migration Risk Analytics | *Python, scikit-learn, React* | GitHub

Aug. – Sept. 2026

- Analyzed 1,405,496 CPS survey records from 2010–2019 using Python and pandas; performed exploratory data analysis (EDA), missing-value handling, and age and income feature engineering.
- Compared logistic regression with 2 histogram gradient-boosting models using temporal validation; excluded target-derived predictors and removed 88,701 overlapping respondent records from earlier partitions.
- Selected gradient boosting and a classification threshold on validation data, achieving ROC-AUC 0.6994 and average precision 0.0475 on 264,173 test records, a 21.1% relative gain in average precision over logistic regression.

VoiceViz | *React, TypeScript, Supabase, SQL* | GitHub

Feb. – Mar. 2026

- Developed a Generative AI analytics prototype that lets users ask data questions by voice or text, generating schema-aware SQL and plain-language explanations.
- Integrated 2 database systems, PostgreSQL and MySQL, with connection testing, schema inspection, and SELECT-statement checks before query execution.
- Built interactive charts, schema browsing, and query-history interfaces for exploring results, with authentication and protected routes.

Steg-Drop | *Python, FastAPI, OpenCV, NumPy* | GitHub

Mar. – Apr. 2026

- Developed a text and file hiding workflow using AES-256-GCM encryption, PBKDF2 key derivation, and authentication-tag verification during recovery.
- Implemented edge-adaptive image steganography with OpenCV Canny detection, payload capacity checks, and lossless PNG output.
- Built 2 FastAPI endpoints for encoding and decoding, integrating a browser interface with input validation and timing metrics.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, HTML, CSS

ML & Analytics: scikit-learn, pandas, NumPy, Matplotlib, Seaborn, OpenCV

Methods: Exploratory Data Analysis (EDA), Feature Engineering, Classification, Model Evaluation

Generative AI: Ollama, RAG, Hybrid Retrieval, NLP, LLM Integration, Prompt Engineering

Engineering: FastAPI, React, Supabase, PostgreSQL, MySQL, REST APIs

CERTIFICATIONS & TRAINING

IBM SkillsBuild — Artificial Intelligence & Data Analytics

IBM

2026

[View Certificate](#) | [Credly Badges](#)

- Completed: [Make Agentic AI Work for You](#) | [Generative AI Essentials: Using LLMs](#) | [Getting Started with Data | Data Fundamentals](#)
- Topics: Generative AI, LLMs, Prompt Engineering, IBM Granite, RAG, AI Agents, Multi-Agent Systems, Data Summarization & Classification, Data Visualization, Cloud Computing, ML, Python

ACHIEVEMENTS

JEE Main 2023 — AIR 5427 among 1.2+ million candidates nationwide